TEMPEST Line of Sight to Magnetic Target Fusion

Regime-adaptive nonequilibrium closures for magnetically driven fusion systems

Five-year hypothesis

Develop non-local heat-flux closures beyond Braginskii that preserve the trusted MHD limit while adding evolved nonequilibrium moments where magnetized transport becomes non-local.

Start with the drive physics, then locate the closure losses that matter for optimal design.

particle/MD
many-body correlations

kinetic
non-Maxwellian transport

fluid/MHD
moments, EOS, opacity

turbulence/target
mix, MRTI, yield

Triple closure as the organizing lens

The central scientific question and testable hypothesis

The closure must add nonequilibrium information without giving up the reliability of the radiation-MHD carrier.

Scientific question

What is the smallest evolved nonequilibrium state that captures the transport and species-memory effects relevant to a magnetized implosion, while preserving a well-posed fluid/MHD system and recovering trusted equilibrium limits?

Hypothesis

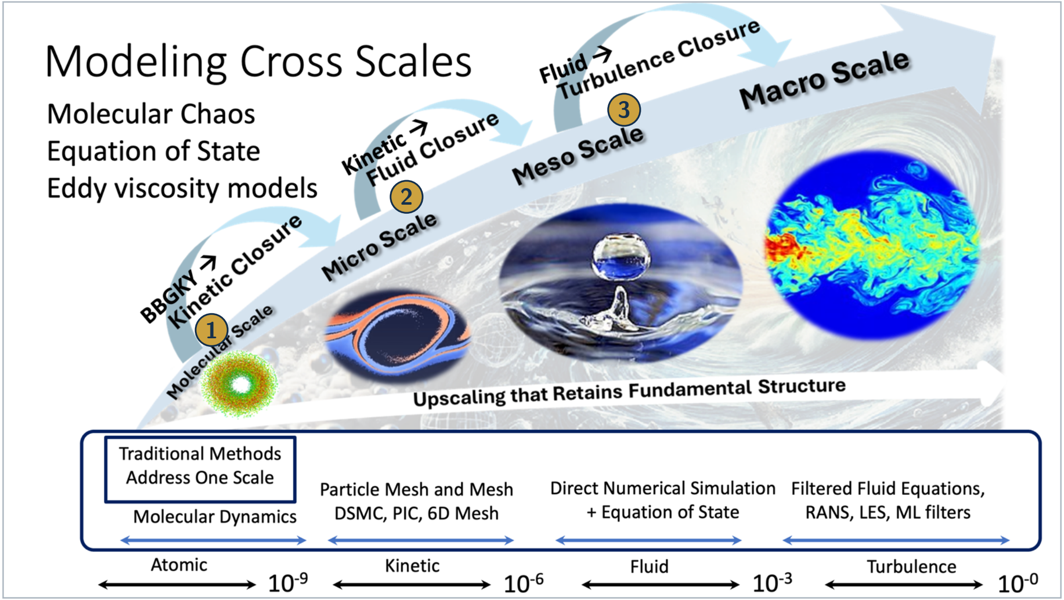
1. A finite hierarchy of moments can represent the dominant departures from LTE/Maxwellian behavior in selected MTF regimes.
2. Structure-preserving SciML can learn the unclosed fluxes and collision/relaxation operators from low-rank kinetic and MD data.
3. A regime indicator can blend the learned closure with standard tables, improving accuracy without sacrificing robust equilibrium performance.

Falsifiable tests

- [ ] Does a compact moment state predict held-out kinetic observables?
- [ ] Does it preserve hyperbolicity, positivity, entropy production, and conservation?
- [ ] Does it recover EOS, diffusion, and Braginskii limits as Knudsen/correlation parameters vanish?
- [ ] Does it improve an integrated observable that matters to target design?

physics fidelity | **mathematical safety** | **computational tractability** | **application impact**

Sources: (TEMPEST Team 2025; Huang et al. 2023; Christlieb et al. 2025).

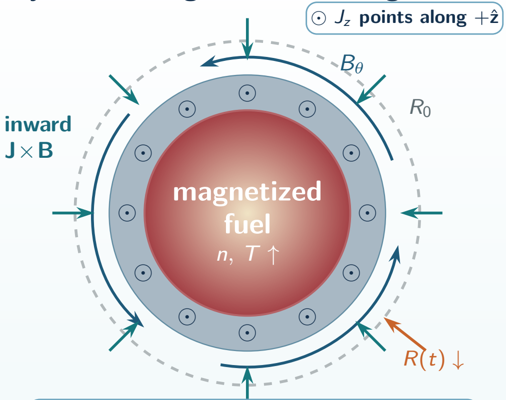


Triple closure lens: $U = Pu^{\varepsilon}$, $\partial_t U = \mathcal{F}_{\text{resolved}}(U) + \mathcal{C}_{\text{lost}}(U, \nabla U, \text{history}; \mu)$ $\mathcal{C}_{\text{lost}} = \mathcal{C}_{\text{BBGKY}\to\text{kinetic}} \oplus \mathcal{C}_{\text{kinetic}\to\text{fluid}} \oplus \mathcal{C}_{\text{fluid}\to\text{turbulence}}$: learn what is lost without breaking conservation, entropy, admissibility, or memory.

Magnetized target fusion: the electromagnetic drive

An axial current generates an azimuthal field and a radially inward $\mathbf{J}\times\mathbf{B}$ compression force.

Cylindrical target, viewed along $+\hat{\mathbf{z}}$



During one implosion, density, temperature, collisionality, and ionization all change.

The electromagnetic drive

1. **Current and field geometry**

$$\mathbf{J} = J_z(r,t)\hat{\mathbf{z}}, \qquad \mathbf{B} = B_\theta(r,t)\hat{\boldsymbol{\theta}} + B_z\hat{\mathbf{z}}.$$

$$\mathbf{f}_L = \mathbf{J}\times\mathbf{B} = -J_z B_\theta \hat{\mathbf{r}}$$

2. **Ampère's law creates the azimuthal field**

$$\frac{1}{r}\frac{\partial}{\partial r}(rB_\theta) = \mu_0 J_z, \qquad B_\theta(r) = \frac{\mu_0 I_{\mathrm{enc}}(r)}{2\pi r}.$$

3. **Magnetic pressure and curvature both compress**

$$f_r = -\frac{\partial}{\partial r}\left(\frac{B_\theta^2}{2\mu_0}\right) - \frac{B_\theta^2}{\mu_0 r} < 0.$$

current J_z → θ field B_θ → force $f_r < 0$

The physics closure cannot be fixed once and for all: it must remain accurate as the plasma moves between regimes.

Literature: Lindemuth & Kirkpatrick, *Nucl. Fusion* 23 (1983); Slutz et al., *Phys. Plasmas* 17 (2010); Gomez et al., *Phys. Rev. Lett.* 113 (2014).

What “line of sight” means for a TEMPEST research activity

A common Center practice for connecting fundamental science to a consequential application decision.

A line of sight is a traceable chain from fundamental science to an application decision.

A project need not simulate the entire application—here, a fusion target.
It must identify the improved assumption, shared asset, validation path, and partner decision.

1. **Application question** — What limits predictive target design? →
2. **Center science** — Which closure or scale bridge? →
3. **Shared asset** — Code, data, benchmark, or theory →
4. **Validation ladder** — Kinetic $\longrightarrow$ experiment $\longrightarrow$ target →
5. **Partner outcome** — Design, uncertainty, and workforce

Every activity should be able to answer

- Which non-canonical regime is being addressed?
- Which level of the triple-closure hierarchy is changed?
- What measurable application quantity could improve?

Every activity should return to the Center

- A reusable “Center artifact” with provenance and uncertainty;
- a cross-pillar collaboration and trainee pathway;
- a milestone decision: continue, pivot, merge, or stop.

From cross-cutting science to an explicit target example

A 10-year line of sight is based on the evidence in hand. TEMPEST research will open doors to new questions; as the Center learns, the line of sight should be revisited and adapted rather than treated as fixed.

Sources: (TEMPEST Team 2025; TEMPEST Team 2026).

State of the art: integrated radiation hydrodynamics / radiation MHD

Baseline: MHD-scale fields plus tabulated EOS/opacity, radiation transport, and reduced plasma-transport closures.

Predictive ICF and pulsed-power target design uses an integrated reduced-physics system.
The current baseline couples shock-capturing radiation hydrodynamics or radiation MHD to tabulated equations of state and opacity, multigroup radiation transport, and reduced plasma-transport packages.

Resolved fields

- mass, species fractions, and momentum;
- ion, electron, and radiation energies;
- magnetic field, shocks, interfaces, MRTI, and mix.

Constitutive closure

- EOS, ionization, opacity, emissivity;
- thermal/electrical transport, viscosity, diffusion;
- equilibration, mixture rules, reduced kinetic corrections.

Integrated packages

- multigroup radiation diffusion;
- anisotropic conduction, extended-MHD/Nernst effects;
- burn, alpha deposition, laser/circuit models, diagnostics.

Pacific Fusion / FLASH baseline

Three-temperature extended MHD with anisotropic heat conduction, electron–ion equilibration, multigroup radiation diffusion, tabulated multi-species EOS and opacities, extended-MHD effects, driver models, and diagnostics. The published benchmarks assume each species is in local thermodynamic equilibrium, use a partial-pressure mixture approximation, and draw tabular EOS data from SESAME and PROPACEOS.

This is the modeling stack to augment: learn only missing closure corrections, while preserving the integrated radiation-MHD carrier.

Representative state-of-the-art references

Ellison et al., "Validation of FLASH for Magnetically Driven Inertial Confinement Fusion Target Design," *Phys. Plasmas* 32, 093902 (2025); Marinak et al., "How Numerical Simulations Helped to Achieve Breakeven on the National Ignition Facility," *Phys. Plasmas* 31, 070501 (2024); Haines, "Charged Particle Transport Coefficient Challenges in High Energy Density Plasmas," *Phys. Plasmas* 31, 050501 (2024); Michta et al., "Code-to-Code Comparison between FLASH and HYDRA in Gas-Puff Z-Pinch Simulations," *Phys. Plasmas* 31, 122703 (2024).

Sources: Ellison et al. (2025); Haines (2024); Marinak et al. (2024); Michta et al. (2024).

Suggested missing physics from the literature for this regime

Five-year Opportunity: augment equilibrium EOS, opacity, and transport with evolved nonequilibrium moments and structure-preserving SciML closures.

For MTF, augmenting equilibrium EOS, opacity, and transport tables with evolved nonequilibrium moments and structure-preserving SciML closures should by Year 5 enable at least a 20% improvement in an uncertainty-aware optimal-design objective for magnetically driven fusion systems.

Missing physics the closure hierarchy must represent

- shock-driven multi-ion stratification, species drift, and partial pressures;
- ion viscous heating and the full anisotropic magnetized stress;
- nonlocal electron/ion transport, kinetic tails, and history dependence;
- heterogeneous warm-dense mixtures, many-body correlations, composition effects, quantum degeneracy, non-Maxwellian structure, and coupled radiation/material nonequilibrium.

Taitano, Simakov, Chacón, and Keenan (*Phys. Plasmas*, 2018): *Yield Degradation in ICF Implosions Due to Shock-Driven Kinetic Fuel-Species Stratification and Viscous Heating.*
Missing physics: shock-imprinted fuel-species stratification and ion viscous heating. **Reported effect:** approximately 10–15% yield variation from each mechanism.

Sam et al. (arXiv:2604.21149, 2026): *Development of Anisotropic Magnetized Viscosity for Magnetized Liner Inertial Fusion Simulations in FLASH.*
Missing physics: the full Braginskii magnetized viscosity tensor, vortical dissipation, and MRTI stabilization.
Reported effect: up to approximately 134% higher yield than the inviscid calculation at the largest seeded perturbation.

Rinderknecht, Amendt, Wilks, and Collins (*PPCF*, 2018): *Kinetic Physics in ICF: Present Understanding and Future Directions.* **Missing physics:** long-mean-free-path kinetics, nonlocal transport, multi-ion separation, self-generated fields, and kinetic interfaces.

Stanton, Glosli, and Murillo (*Phys. Rev. X*, 2018); **Padilla-Gomez et al.** (*Phys. Plasmas*, 2026). **Missing physics:** heterogeneous many-body correlations, strong coupling, quantum statistics, and non-Maxwellian warm-dense dynamics.

Sources: Taitano et al. (2018); Sam et al. (2026); Rinderknecht et al. (2018); Stanton et al. (2018); Padilla-Gomez et al. (2026).

Braginskii heat flux in MHD: the trusted local limit

Braginskii is the asymptotic anchor; TEMPEST learns the non-local correction when that limit fails.

Do not replace Braginskii everywhere: preserve it as the collisional, near-Maxwellian, small-Knudsen-number limit and add evolved moments only where the plasma is non-local.

Magnetized heat flux closure

$$\mathbf{q}_s^{\mathrm{Brag}} = -\kappa_{\parallel s}\,\mathbf{b}(\mathbf{b}\cdot\nabla T_s) - \kappa_{\perp s}(\mathbf{I}-\mathbf{bb})\nabla T_s - \kappa_{\wedge s}\,\mathbf{b}\times\nabla T_s$$

$\mathbf{b} = \mathbf{B}/|\mathbf{B}|$; coefficients depend locally on n_s, T_s, Z, $\omega_{cs}\tau_s$.

parallel conduction ⊕ cross-field conduction ⊕ Righi–Leduc heat flow

Where it enters radiation MHD

$$\partial_t E_s + \nabla\cdot\Big[(E_s + p_s)\mathbf{u}_s + \mathbf{q}_s + \mathbf{\Pi}_s\mathbf{u}_s\Big] = Q_s + R_s$$

The heat flux sets preheat, pressure balance, temperature separation, transport into the liner/fuel interface, and the energy available for compression and instability growth.

Assumptions

local gradient law; collisional Chapman–Enskog ordering; near Maxwellian; $\lambda_{\mathrm{mfp}} \ll L_T$

Breakdown indicators

heat-front memory; kinetic tails; strong gradients; shocks; species separation; evolving magnetization

TEMPEST target

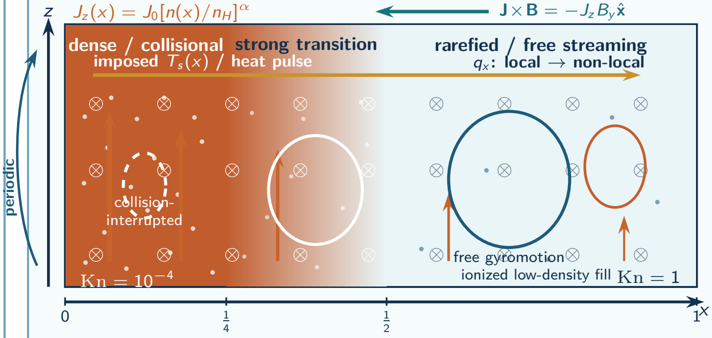
learn $\mathbf{q}_s = \mathbf{q}_s^{\mathrm{Brag}} + \Delta\mathbf{q}_s^{\mathrm{NL}}$ with conservation, hyperbolicity, positivity, entropy, and fallback

Sources: (Braginskii 1965; Haines 2024; Sam et al. 2026).

Plasma slab test problem: a 1D2V cross-regime benchmark

A controlled benchmark spans the Braginskii limit, non-local transition physics, and free streaming.

Physical-space embedding in the x–z plane



$\otimes$ B_y: field into page $\uparrow$ J_z: density-dependent current → q_x: heat transport

Model and closure test

1D2V kinetic geometry

$f_s = f_s(x, v_x, v_z, t)$, $\partial_z = 0$
$\mathbf{B} = B_y(x,t)\hat{\mathbf{y}}$, $\mathbf{J} = J_z(x,t)\hat{\mathbf{z}}$
$\partial_x B_y = \mu_0 J_z$, $\mathbf{f}_L = -J_z B_y \hat{\mathbf{x}}$

Controlled profiles

- quasi-neutral: $n_i = n_e = n(x)$
- $n/n_H : 1 \to \eta$ over $x \in [1/4, 1/2]$
- $J_z = J_0(n/n_H)^\alpha$, $\mathrm{Kn} : 10^{-4} \to 1$
- prescribed $T_s(x)$ or a boundary heat pulse

Three-model comparison

Kinetic reference	q_x^{kin} and held-out moments
Braginskii MHD	q_x^{Brag}: trusted local limit
Grad–SciML	$q_x^{\mathrm{Brag}} + \Delta q_x^{\mathrm{NL}}$

▶ dense: correlations / EOS / transport
▶ transition: memory / nonlocality
▶ dilute: kinetic tails / free streaming

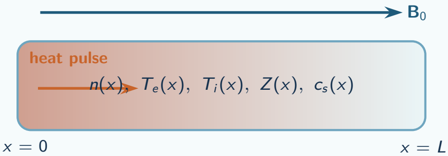
One benchmark links BBGKY → kinetic data to kinetic → Grad/MHD closure; multi-D extensions add fluid → turbulence.

Sources: Braginskii (1965); Baalrud & Daligault, *Phys. Rev. Lett.* (2013); Crestetto, Deluzet & Doyen, *J. Plasma Phys.* (2020).

Plasma slab test problem: isolate non-local heat flux first

A reduced slab creates a publishable, shareable benchmark before inserting the closure into full MTF target design.

Canonical slab geometry



Drive either a boundary heat pulse or a localized volumetric deposition through a magnetized, stratified plasma with a controllable density/composition gradient.

The slab is the interface between kinetic truth data, Braginskii MHD, and the learned Grad closure.

Three-model comparison

Kinetic reference Vlasov/Fokker–Planck, low-rank kinetic, or particle data in strongly magnetized regimes

MHD baseline two-temperature MHD with Braginskii heat flux and existing transport tables

Learned closure Grad-moment state with $\Delta\mathbf{q}^{\mathrm{NL}}$ trained to kinetic data and constrained by the Braginskii limit

What the benchmark measures

Regime coordinates: $\mathrm{Kn} = \lambda_{\mathrm{mfp}}/L_T$, $\chi = \omega_c\tau$, β, Mach number, density-gradient scale, composition, ionization state.

Observables: heat-front speed, non-local preheat, q/q_{Brag}, electron–ion temperature separation, pressure anisotropy, entropy production, and energy conservation.

Center artifact: an open slab input deck, kinetic snapshots, uncertainty tags, and a closure-test notebook usable by FLASH, Parthenon, and Pacific Fusion partners.

Sources: (Braginskii 1965; Haines 2024; Ellison et al. 2025; TEMPEST Team 2025).

Five-year plan: non-local heat flux beyond Braginskii

Start with hyperbolic ML moment closures, move through Grad and two-fluid MHD, and validate against kinetic data, slab tests, and experiment.

Modeling, simulation, and experiment: Sean Couch, Qi Tang, Brian O'Shea, Andrew Christlieb, Lei, Zayernouri, Shang, Murillo, Stanton, Li, Gilbert, Pacific Fusion, and TEMPEST trainees/partners, and ??? Open to Collaborators.

Year 1 — **Grad closure foundation**
Extend Huang–Cheng–Christlieb hyperbolic ML moment closure ideas from RTE to Grad; define the slab, admissible set, and Braginskii recovery tests.

Year 2 — **Grad $\rightarrow$ two-fluid MHD**
Derive the moment-to-MHD reduction; train adaptive heat-flux, stress, and relaxation closures on Braginskii-limit data plus strongly magnetized kinetic data.

Year 3 — **2D Grad-MHD extension**
Adapt Huang's 2D symmetrizable-hyperbolic RTE program to Grad-MHD; couple low-rank kinetic training data to the MHD solver and regime indicator.

Year 4 — **Verification and experiment**
Verify against held-out kinetic data and the plasma slab experiment; quantify uncertainty, robustness, fallback to Braginskii, and transport-observable improvement.

Year 5 — **Partner design loop**
Deliver a vetted closure module, slab benchmark, and Pacific-Fusion-relevant design study that reports closure-driven improvement in predictive uncertainty or optimal design.

Sources: (Huang et al. 2023; Huang 2026; Christlieb et al. 2025; Kahza, Taitano, et al. 2024; Kahza, Qiu, et al. 2025; Zhao et al. 2025).

Cooperative Agreement alignment I: topics a–c

The non-local heat-flux project is a concrete example of line of sight from fundamental closure science to MTF partner use.

a

Unifying intellectual theme and objectives

Use non-local heat flux beyond Braginskii as a focused kinetic-to-MHD closure problem inside the TEMPEST triple-closure hierarchy. It connects particle/kinetic physics, Grad moments, MHD/turbulence, experiment, and MTF design.

b

Coherent research goals and planned activities

Derive the Grad/two-fluid/MHD hierarchy; construct structure-preserving SciML closures for heat flux and relaxation; train on Braginskii-limit and strongly magnetized kinetic data; test in the slab and then in integrated MHD benchmarks.

c

Education and knowledge transfer integrated with research

Each trainee produces a Center artifact: slab input deck, kinetic dataset, closure module, validation notebook, or documentation. Pacific Fusion engagement provides design relevance; cross-pillar mentoring links modeling, simulation, and experiment.

Success measure: a reusable, verified closure workflow that teaches the Center how to move from cross-cutting science to a target-application decision.

Sources: (TEMPEST Team 2025; TEMPEST Team 2026; Pacific Fusion 2026).

Cooperative Agreement alignment II: topics d–g

Infrastructure, management, boldness, and continuity are built into the work package rather than added afterward.

d Shared infrastructure

Open slab benchmark; kinetic snapshots; uncertainty metadata; HDF5/NetCDF formats; CI-tested closure kernels; containers; DOI release through TEMPEST data/software infrastructure.

e Leadership and partner integration

Named modeling/simulation/experiment team; monthly milestone checks; Pacific Fusion partner review; work package mapped to Center artifacts and resource decisions.

f Bold research, opportunities, and stop rules

High-risk step: replace local heat-flux closure by a constrained non-local Grad/SciML state. Continue only if it passes invariants, Braginskii limits, held-out kinetic tests, and slab validation; otherwise pivot moment set, training data, or regime indicator.

g Continuity if personnel change

Two-code path: standalone slab plus FLASH/Parthenon-style interface. Each component has documented handoffs, archived datasets, reproducible notebooks, and at least two trained users across institutions.

Sources: (TEMPEST Team 2025; TEMPEST Team 2026).